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Subject: JAVA
Assignment:1
Form Number: 251101107

1. Calculate the area of a rectangle given its length and width.

Solution:

```
public class area_of_rectangle{
        public static void main(String[] args){
        int length = Integer.parseInt(args[0]);
        int breadth = Integer.parseInt(args[1]);
        System.out.println("Area of rectangle:"+ length * breadth);
        }
}
```

```
C:\Users\Asus\OneDrive\Desktop\Java_programs>javac area_of_rectangle.java

C:\Users\Asus\OneDrive\Desktop\Java_programs>java area_of_rectangle 8 9
Area of rectangle:72
```

2. Convert a temperature from Celsius to Fahrenheit using the formula: F = (C * 9/5) + 32.

Solution:

```
class celsius_to_fahrenheit{
        public static void main(String []args){
   int celsius = Integer.parseInt(args[0]);
        System.out.println("Celsius to Fahrenheit: "+ ((celsius * 9/5) + 32));

        }
}
```

```java
class celsius_to_fahrenheit{
    public static void main(String []args){
    int celsius = Integer.parseInt(args[0]);
    System.out.println("Celsius to Fahrenheit: "+ ((celsius * 9/5) + 32));

    }
}

```

3. Calculate the area of a circle given its radius using the formula: A = π * r^2.

Solution:

```
class area_of_circle{
        public static void main(String [] args){
                double r = Double.parseDouble(args[0]);
                System.out.println("Area_of_Circle: "+ (Math.PI * r * r));
        }
}
```

```java
// 3.Calculate the area of a circle given its radius using the formula: A = π * r^2.

class area_of_circle{
    public static void main(String [] args){
        double r = Double.parseDouble(args[0]);
        System.out.println("Area_of_Circle: "+ (Math.PI * r * r));
    }
}
```

4. Compute the hypotenuse of a right triangle using the Pythagorean theorem: c = sqrt(a^2 + b^2).

Solution:

```
class right_triangle{
        public static void main(String [] args){
                int a = Integer.parseInt(args[0]);
                int b = Integer.parseInt(args[1]);

                System.out.println("The Hypotenuse of right_triangle:"+ Math.sqrt(a*a + b*b));
        }
}
```

```java
// 4.Compute the hypotenuse of a right triangle using the Pythagorean theorem: c = sqrt(a^2 + b^2).

class right_triangle{
    public static void main(String [] args){
        int a = Integer.parseInt(args[0]);
        int b = Integer.parseInt(args[1]);

        System.out.println("The Hypotenuse of right_triangle:"+ Math.sqrt(a*a + b*b));
    }
}
```

5. Calculate the simple interest on an investment using the formula: A = P * T * R / 100;

Solution:

```java
class simple_interest{
        public static void main(String []args){
        int P = Integer.parseInt(args[0]);
        int T = Integer.parseInt(args[1]);
        int R = Integer.parseInt(args[2]);

        System.out.println("The simple_interest is:"+ ((P * T * R)/100));
        }
}
```

```java
// 5.Calculate the simple interest on an investment using the formula: A = P * T * R / 100;

class simple_interest{
    public static void main(String []args){
    int P = Integer.parseInt(args[0]);
    int T = Integer.parseInt(args[1]);
    int R = Integer.parseInt(args[2]);

    System.out.println("The simple_interest is:"+ ((P * T * R)/100));
    }
}
```

6. Calculate the compound interest on an investment using the formula: A = P * (1 + r/n)^(n*t).

Solution:

```java
class compound_interest{
        public static void main(String [] args){

                double p = Double.parseDouble(args[0]);
                double r = Double.parseDouble(args[1]);
                double n = Double.parseDouble(args[2]);
```

```
            double t = Double.parseDouble(args[3]);

            double amount = p * Math.pow((1 + r/n), (n*t));

            System.out.println("The compound interest: " + amount);
        }
}
```

```
// 6.Calculate the compound interest on an investment using the formula: A = P * (1 + r/n)^(n*t).

class compound_interest{
    public static void main(String [] args){

        double p = Double.parseDouble(args[0]);
        double r = Double.parseDouble(args[1]);
        double n = Double.parseDouble(args[2]);
        double t = Double.parseDouble(args[3]);

        double amount = p * Math.pow((1 + r/n), (n*t));

        System.out.println("The compound interest: " + amount);
    }
}
```

7. Determine the volume of a sphere given its radius using the formula: V = (4/3) * π * r^3.

Solution:

```
class volume_of_sphere{
        public static void main(String []args){
            int r = Integer.parseInt(args[0]);

            double v = (4.0/3) * 3.14 * r*r*r;
            System.out.println("The volume of sphere:"+ v);
        }
}
```

```
// 7.Determine the volume of a sphere given its radius using the formula: V = (4/3) * π * r^3.

class volume_of_sphere{
    public static void main(String []args){
        int r = Integer.parseInt(args[0]);

        double v = (4.0/3) * 3.14 * r*r*r;
        System.out.println("The volume of sphere:"+ v);
    }
}
```

8. Convert a distance from kilometers to miles using the formula: miles = kilometers * 0.621371.

Solution:

```
class kilometers_to_miles{
        public static void main(String [] args){
                int kilometers = Integer.parseInt(args[0]);
                double miles = kilometers * 0.621371;

                System.out.println("kilometers to miles:" + miles);
        }
}
```

```java
// 8.Convert a distance from kilometers to miles using the formula: miles = kilometers * 0.621371.

class kilometers_to_miles{
    public static void main(String [] args){
        int kilometers = Integer.parseInt(args[0]);
        double miles = kilometers * 0.621371;

        System.out.println("kilometers to miles:" + miles);
    }
}
```

9. Calculate the sum of the first n natural numbers using the formula: sum = (n * (n + 1)) / 2.

Solution:

```
class sum{
        public static void main(String [] args){
                int n = Integer.parseInt(args[0]);
                int sum = (n * (n + 1)) / 2;

                System.out.println("The sum of " + n + "natural number is:"+sum);

        }
}
```

```java
// 9.Calculate the sum of the first n natural numbers using the formula: sum = (n * (n + 1)) / 2.

class sum{
    public static void main(String [] args){
        int n = Integer.parseInt(args[0]);
        int sum = (n * (n + 1)) / 2;

        System.out.println("The sum of " + n + "natural number is:"+sum);

    }
}
```

10. Compute the area of a triangle given the lengths of its three sides using Heron's formula.

Solution:

```
class area_of_triangle{
        public static void main(String [] args){
                double a = Double.parseDouble(args[0]);
                double b = Double.parseDouble(args[1]);
                double c = Double.parseDouble(args[2]);
                double s = (a + b + c) /2;
                double A = Math.sqrt(s * (s - a) * (s - b) * (s - c));


                System.out.println("The semi perimeter:"+ s);
                System.out.println("The area of triangle:"+ A);
        }
}
```

```java
// 10.Compute the area of a triangle given the lengths of its three sides using Heron's formula.

class area_of_triangle{
    public static void main(String [] args){
        double a = Double.parseDouble(args[0]);
        double b = Double.parseDouble(args[1]);
        double c = Double.parseDouble(args[2]);
        double s = (a + b + c) /2;
        double A = Math.sqrt(s * (s - a) * (s - b) * (s - c));


        System.out.println("The semi perimeter:"+ s);
        System.out.println("The area of triangle:"+ A);
    }
}
```